Blood Bank Management System - Complete Test Cases

Total: 160 Test Cases (16 FRs × 10 tests each)

FR-01: User Registration

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
1-001	User Registration	Register with valid data	Access registration page	1. Enter email, password, name, contact 2. Accept terms 3. Submit	Email: user1@test.com, Password: Test@123, Name: John, Contact: 9876543210	Account created with unique user ID		
1-002	User Registration	Register with existing email	User with email exists	1. Enter existing email 2. Enter password 3. Submit	Email: existing@test.com	Error: Email already registered		
1-003	User Registration	Register with invalid email	Registration page open	1. Enter invalid email 2. Submit	Email: invalidemail	Error: Invalid email format		
1-004	User Registration	Register with weak password	Registration page open	1. Enter weak password 2. Submit	Password: 123	Error: Password does not meet requirements		
1-005	User Registration	Register with missing required fields	Registration page open	1. Leave required field empty 2. Submit	Missing: Contact number	Error: All required fields mandatory		
1-006	User Registration	Register with special characters in name	Registration page open	1. Enter name with special chars 2. Submit	Name: John@#$	Error: Invalid characters in name		
1-007	User Registration	Register with very long input	Registration page open	1. Enter name >255 chars 2. Submit	Name: >255 characters	Error: Input exceeds maximum length		

1-008	User Registration	Register with SQL injection attempt	Registration page open	1. Enter SQL injection in name 2. Submit	Name: '; DROP TABLE users;--	Input sanitized, account created safely		
1-009	User Registration	Register with duplicate phone number	User with phone exists	1. Enter existing phone 2. Submit	Phone: 9876543210	Warning: Phone already registered		
1-010	User Registration	Verify email confirmation	Registration submitted	1. Check email 2. Click confirmation link	Email account created	Email verified, account fully activated		

FR-02: User Login

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
2-001	User Login	Login with valid credentials	User account exists	1. Navigate to login 2. Enter email and password 3. Click Login	Email: donor@test.com, Password: Pass@123	Login successful, redirected to dashboard		
2-002	User Login	Login with invalid password	User account exists	1. Enter email 2. Enter wrong password 3. Submit	Password: WrongPass	Error: Invalid email or password		
2-003	User Login	Login with non-existent email	No preconditions	1. Enter non-existent email 2. Submit	Email: nonexistent@test.com	Error: Invalid email or password		
2-004	User Login	Login account lockout after 5 attempts	User account exists	1. Attempt login 5 times with wrong password 2. Try login with correct	5 failed attempts	Account locked after 5th attempt		

				password				
2-005	User Login	Session timeout after inactivity	User logged in	1. Leave browser idle for 30 min 2. Try to perform action	Inactivity : 30 minutes	Session expired, redirected to login		
2-006	User Login	Login with empty email field	Login page open	1. Leave email empty 2. Enter password 3. Submit	Email: empty	Error: Email is required		
2-007	User Login	Login with empty password field	Login page open	1. Enter email 2. Leave password empty 3. Submit	Password : empty	Error: Password is required		
2-008	User Login	Login with SQL injection	Login page open	1. Enter SQL injection in email 2. Submit	Email: ' OR 1=1--	Input sanitized, normal login flow		
2-009	User Login	Verify secure session cookie	User logged in	1. Login 2. Check browser dev tools 3. Verify cookie properties	Check HttpOnly, Secure, SameSite flags	Cookie has HttpOnly, Secure, SameSite attributes		
2-010	User Login	Login from different browsers simultaneously	Multiple browsers available	1. Login in Browser 1 2. Login in Browser 2 3. Perform action in Browser 1	Two browser sessions	Both sessions work independently		

FR-03: Donor Registration

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
3-001	Donor	Register	Staff	1.	Name:	Donor		

	Registration	donor with valid data	logged in	Navigate to Donor Registration 2. Enter all fields 3. Submit	John Doe, Blood: O+, Contact: 9876543210	created with unique Donor ID		
3-002	Donor Registration	Register donor with duplicate ID	Donor with ID exists	1. Try to assign existing Donor ID 2. Submit	Donor ID: D001	Error: Donor ID already exists		
3-003	Donor Registration	Register donor with missing name	Registration form open	1. Leave name field empty 2. Enter other fields 3. Submit	Name: empty	Error: Donor name is required		
3-004	Donor Registration	Register donor with invalid blood group	Registration form open	1. Enter invalid blood group 2. Submit	Blood Group: XYZ	Error: Invalid blood group		
3-005	Donor Registration	Register donor with invalid contact	Registration form open	1. Enter non-numeric contact 2. Submit	Contact: ABC123	Error: Contact must be numeric		
3-006	Donor Registration	Register donor with duplicate contact	Donor with contact exists	1. Enter existing contact 2. Submit	Contact: 9876543210	Warning: Contact already registered		
3-007	Donor Registration	Non-staff user tries to register donor	Donor user logged in	1. Try to access Donor Registration 2. Submit form	User Role: Donor	Access denied: Insufficient permissions		
3-008	Donor Registration	Register donor with very long name	Registration form open	1. Enter name >100 chars 2. Submit	Name: 150 characters	Error: Name exceeds maximum length		
3-009	Donor Registrati	Donor registrati	Staff registered	1. Register	Donor ID: D001	Audit log shows		

	on	on history tracking	new donor	donor 2. Check audit log 3. Verify entry		registration with staff ID, timestamp		
3-010	Donor Registration	Register donor and verify data persists	Donor registered	1. Register donor 2. Search for donor 3. View details	Donor ID: D100	All entered data matches persisted records		

FR-04: Donor Management

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
4-001	Donor Management	Update donor contact information	Donor exists	1. Navigate to Donor Search 2. Find donor 3. Update contact 4. Save	Donor ID: D001, New Contact: 9999999999	Contact updated, change logged		
4-002	Donor Management	Search donor by ID	Donor exists	1. Go to Donor Search 2. Enter Donor ID 3. Search	Donor ID: D001	Donor details displayed		
4-003	Donor Management	Search donor by name	Donor exists	1. Go to Donor Search 2. Enter donor name 3. Search	Name: John	All donors with matching name displayed		
4-004	Donor Management	View donor details	Donor found	1. Search for donor 2. Click on donor record 3. View all details	Donor ID: D001	All donor info displayed including history		
4-005	Donor Management	Donor edits own profile	Donor logged in	1. Navigate to My Profile 2.	New Contact: 8888888888	Profile updated, cannot change		

				Update contact 3. Save		blood group		
4-006	Donor Management	Add multiple donor records	Staff logged in	1. Register donor 1 2. Register donor 2 3. Register donor 3	3 different donors	All 3 donors added with unique IDs		
4-007	Donor Management	Search returns no results	Donor does not exist	1. Search for non-existent donor 2. View results	Donor ID: D9999	Message: No donors found		
4-008	Donor Management	Unauthorized user tries to update donor	Donor user logged in	1. Try to access update form 2. Attempt to update	User: Donor role	Access denied		
4-009	Donor Management	Update invalid field values	Donor edit open	1. Update with invalid data 2. Save	Contact: ABC, Blood: XYZ	Error: Invalid values not saved		
4-010	Donor Management	Verify audit trail for updates	Donor record updated	1. Update donor 2. Check audit log 3. Verify entry	Change: Contact number	Audit shows user, action, timestamp, old/new values		

FR-05: Blood Donation Recording

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
5-001	Blood Donation Recording	Record donation with valid data	Staff logged in, donor exists	1. Navigate to Record Donation 2. Enter all fields 3. Submit	Donor: D001, Blood: O+, Qty: 450ml, Date: 2024-01-15	Donation saved with unique Donation ID		

5-002	Blood Donation Recording	Record donation with future date	Recording form open	1. Enter future date 2. Submit	Date: 2025-12-31	Error: Future dates not allowed		
5-003	Blood Donation Recording	Record donation with zero quantity	Recording form open	1. Enter 0 ml 2. Submit	Quantity: 0 ml	Error: Quantity must be > 0		
5-004	Blood Donation Recording	Record donation with negative quantity	Recording form open	1. Enter -100 ml 2. Submit	Quantity: -100 ml	Error: Quantity cannot be negative		
5-005	Blood Donation Recording	Record duplicate donation same day	Donor already donated today	1. Record second donation 2. Submit	Same donor, same date	Error/ Warning: Cannot donate twice same day		
5-006	Blood Donation Recording	Verify donation inventory update	Initial inventory: 2000ml O+	1. Record donation 450ml 2. Check inventory 3. Verify increase	Donation: 450ml O+	Inventory: 2450ml O+		
5-007	Blood Donation Recording	Record donation invalid blood group	Recording form open	1. Select invalid blood group 2. Submit	Blood Group: XYZ	Error: Invalid blood group		
5-008	Blood Donation Recording	Non-staff tries to record donation	Donor user logged in	1. Try to access Record Donation 2. Submit	User: Donor role	Access denied		
5-009	Blood Donation Recording	Verify donation audit trail	Donation recorded	1. Record donation 2. Check audit log 3. Verify	Donation ID: DN001	Audit shows staff ID, donor, date, quantity, timestamp		
5-010	Blood Donation Recording	Record donation missing donor ID	Recording form open	1. Leave donor ID empty 2. Submit	Donor ID: empty	Error: Donor ID is required		

FR-06: Blood Bank Management

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
6-001	Blood Bank Management	Admin adds new blood bank	Admin logged in	1. Navigate to Blood Bank Management 2. Click Add 3. Enter all fields 4. Submit	Name: Central Bank, Location: City, Contact: 1234567890	Blood bank created with Bank ID		
6-002	Blood Bank Management	Add blood bank duplicate contact	Bank with contact exists	1. Enter existing contact 2. Submit	Contact: 1234567890	Error: Contact already registered		
6-003	Blood Bank Management	Update blood bank information	Blood bank exists	1. Find bank 2. Edit information 3. Save	Update contact to 9999999999	Bank info updated		
6-004	Blood Bank Management	Deactivate blood bank	Active blood bank exists	1. Select blood bank 2. Click Deactivate 3. Confirm	Bank ID: BB001	Bank status changed to Inactive		
6-005	Blood Bank Management	Non-admin tries to add blood bank	Staff user logged in	1. Try to access Blood Bank Management 2. Try to add	User: Staff role	Access denied		
6-006	Blood Bank Management	Add bank with missing name	Add form open	1. Leave name empty 2. Submit	Name: empty	Error: Bank name is required		
6-007	Blood Bank Management	View all blood banks	Multiple banks exist	1. Navigate to Bank list 2. View all banks	5 banks in system	All 5 banks displayed		

6-008	Blood Bank Management	Search blood bank by name	Banks exist	1. Search by name 2. View results	Search: Central	Matching banks displayed		
6-009	Blood Bank Management	Verify bank audit trail	Bank created	1. Create bank 2. Check audit log	Bank ID: BB001	Audit shows admin ID, action, timestamp		
6-010	Blood Bank Management	Cannot delete bank only deactivate	Bank exists	1. Try to delete bank 2. Look for delete option	Bank ID: BB001	Delete not available, only deactivate		

FR-07: Blood Inventory Management

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
7-001	Blood Inventory Management	Staff views inventory by blood group	Staff logged in	1. Navigate to Inventory 2. Select blood group O+ 3. View	Blood Group: O+	All O+ units displayed with quantities		
7-002	Blood Inventory Management	Staff adds blood to inventory	Staff logged in, O+ inventory : 2000ml	1. Navigate to Inventory Update 2. Select Add 3. Enter 500ml 4. Submit	Add: 500ml O+	Inventory updated to 2500ml		
7-003	Blood Inventory Management	Staff removes blood from inventory	Staff logged in, O+ inventory : 2000ml	1. Select Remove 2. Enter 300ml 3. Provide reason 4. Submit	Remove: 300ml (Transfusion)	Inventory updated to 1700ml		
7-004	Blood Inventory Managem	Cannot remove more	O+ inventory : 500ml	1. Try to remove 600ml 2.	Remove: 600ml	Error: Insufficient		

	ent	than available		Submit		quantity		
7-005	Blood Inventory Management	Filter inventory by blood bank	Multiple banks have inventory	1. Navigate to Inventory 2. Select bank 3. View	Bank: Central Bank	Only that bank inventory displayed		
7-006	Blood Inventory Management	Non-staff cannot modify inventory	Donor logged in	1. Try to add/remove blood 2. Submit	User: Donor	Access denied		
7-007	Blood Inventory Management	Staff can only modify assigned bank	Staff1 assigned to Bank1	1. Try to modify Bank2 inventory 2. Submit	Staff1 accessing Bank2	Access denied or restricted		
7-008	Blood Inventory Management	View inventory all blood groups	Inventory data exists	1. Navigate to Inventory 2. View all blood groups	All 8 blood groups	All groups displayed with quantities		
7-009	Blood Inventory Management	Verify inventory entry audit trail	Inventory updated	1. Update inventory 2. Check audit log	Update: +500ml O+	Audit shows staff, blood group, quantity, reason, timestamp		
7-010	Blood Inventory Management	Search and filter inventory	Multiple records exist	1. Filter by blood group 2. Filter by bank 3. Filter by both	O+ at Central Bank	Results match all selected filters		

FR-08: Automatic Inventory Update

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
8-001	Automatic Inventory	Inventory increases on	Donation recorded	1. Record donation 450ml	Donation: 450ml O+	Inventory increased by 450ml		

	Update	donation		O+ 2. Check inventory 3. Verify increase				
8-002	Automatic Inventory Update	Inventory decreases on request fulfillment	Request fulfilled	1. Fulfill request 500ml A+ 2. Check inventory 3. Verify decrease	Request: 500ml A+	Inventory decreased by 500ml		
8-003	Automatic Inventory Update	Multiple transactions update correctly	Multiple donations recorded	1. Record 3 donations 2. Check each update 3. Verify final quantity	3 donations of different amounts	Inventory reflects sum of all changes		
8-004	Automatic Inventory Update	Concurrent updates handled correctly	Two staff updating simultaneously	1. Staff1 adds 300ml 2. Staff2 adds 200ml 3. Check final inventory	Concurrent: +300ml and +200ml	Inventory shows +500ml correctly		
8-005	Automatic Inventory Update	Inventory timestamp updates	Inventory updated	1. Update inventory 2. Check timestamp 3. Verify accuracy	Update at 14:30:45	Timestamp recorded accurately		
8-006	Automatic Inventory Update	Update history maintained	Multiple updates made	1. Make 5 inventory updates 2. Check history log 3. Verify all entries	5 inventory changes	All changes logged with details		
8-007	Automatic Inventory Update	Rollback if update fails	Update attempted	1. Start update 2. Simulate error	Transaction interrupted	Inventory remains unchanged		

				mid-transaction 3. Check inventory		(rollback)		
8-008	Automatic Inventory Update	Negative inventory prevented	Update attempted	1. Try to reduce below zero 2. Check inventory	Remove more than available	Inventory stays >= 0, update rejected		
8-009	Automatic Inventory Update	Update propagates to search results	Inventory updated	1. Update inventory 2. Search for blood 3. Verify results	Updated: O+ quantity	Search results show updated quantity		
8-010	Automatic Inventory Update	Partial quantity transactions	Inventory available	1. Issue partial quantity 2. Check inventory 3. Verify adjustment	Available: 1000ml, Issue: 750ml, Remain: 250ml	Inventory shows 250ml remaining		

FR-09: Blood Availability Search

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
9-001	Blood Availability Search	Search by blood group only	Blood available	1. Select blood group O+ 2. Search	Blood Group: O+	All O+ units displayed across banks		
9-002	Blood Availability Search	Search by location only	Blood available in locations	1. Select location 2. Search	Location: Bangalore	All banks in location displayed		
9-003	Blood Availability Search	Search by blood group and location	Blood available	1. Select O+ 2. Select location 3. Search	O+ in Bangalore	O+ banks in Bangalore displayed		
9-004	Blood Availability Search	Search with quantity requirement	Blood available	1. Select O+ 2. Enter needed qty:	Need: 500ml O+	Only banks with >= 500ml shown		

				500ml 3. Search				
9-005	Blood Availability Search	Performance: results within 2 seconds	Large database	1. Execute search 2. Measure time	100,000+ records	Results returned < 2 seconds		
9-006	Blood Availability Search	Search returns no results	Blood not available	1. Search rare blood group 2. View results	Rare group in location	Message: No results, suggest nearby banks		
9-007	Blood Availability Search	Sort results by proximity	Multiple results	1. Search for blood 2. Check sort order 3. Verify proximity	User location set	Results sorted by distance		
9-008	Blood Availability Search	Search results include contact info	Results returned	1. Search for blood 2. View result details	O+ search	Bank name, location, contact, hours shown		
9-009	Blood Availability Search	Unauthenticated user can search	Not logged in	1. Access search page 2. Perform search	No login required	Search works without authentication		
9-010	Blood Availability Search	Search multiple blood groups	Search open	1. Select O+, A+, B+ 2. Search	Multiple groups selected	Results for all selected groups		

FR-10: Blood Request

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
10-001	Blood Request	Submit request with valid data	User logged in	1. Navigate to Request Blood 2. Select O+ 3. Enter qty 500ml 4. Set	O+, 500ml, City Hospital, High	Request created with Request ID, status Pending		

				urgency High 5. Submit				
10-002	Blood Request	Submit with insufficient quantity	Only 200ml available	1. Request 500ml 2. Submit	Need: 500ml, Have: 200ml	Request accepted, status Pending, staff notified		
10-003	Blood Request	Submit without blood group	Request form open	1. Leave blood group empty 2. Submit	Blood Group: empty	Error: Blood group required		
10-004	Blood Request	Submit with zero quantity	Request form open	1. Enter quantity 0 2. Submit	Quantity: 0 ml	Error: Quantity must be > 0		
10-005	Blood Request	Submit with missing location	Request form open	1. Leave location empty 2. Submit	Location: empty	Error: Location required		
10-006	Blood Request	Submit with missing contact info	Request form open	1. Leave contact empty 2. Submit	Contact: empty	Error: Contact information required		
10-007	Blood Request	View submitted requests	User submitted request	1. Navigate to My Requests 2. View requests list	Request ID: BR001	All user requests displayed with status		
10-008	Blood Request	Unauthenticated user cannot submit	Not logged in	1. Try to access request form 2. Submit	No authentication	Redirected to login		
10-009	Blood Request	Request with urgent flag	Request form open	1. Select Urgent 2. Submit	Urgency: Critical	Request marked urgent, staff prioritized		
10-010	Blood Request	Verify request audit trail	Request submitted	1. Submit request 2. Check audit log	Request ID: BR001	Audit shows user, blood		

						group, qty, location, timestamp		

FR-11: Request Management

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
11-001	Request Management	Staff views pending requests	Staff logged in, requests exist	1. Navigate to Requests 2. Filter by Pending 3. View list	Status: Pending	All pending requests displayed		
11-002	Request Management	Staff updates request to Accepted	Pending request exists	1. Select request 2. Change status to Accepted 3. Add notes 4. Save	Request ID: BR001	Status updated, user notified		
11-003	Request Management	Staff rejects request with reason	Pending request exists	1. Select request 2. Change to Rejected 3. Enter reason 4. Save	Reason: Blood not available	Status updated, user notified with reason		
11-004	Request Management	Staff marks request as Fulfilled	Accepted request exists	1. Select request 2. Change to Fulfilled 3. Record details 4. Save	Request ID: BR001	Status Fulfilled, inventory decreased		
11-005	Request Management	Non-staff cannot update requests	Donor logged in	1. Try to change request status 2. Submit	User: Donor	Access denied		
11-006	Request Managem	View all requests	Requests with	1. Navigate	All statuses	All requests		

	ent	across statuses	different statuses	to Requests 2. View all		displayed regardless of status		
11-007	Request Management	Filter requests by status	Multiple requests exist	1. Select Accepted status 2. View filtered	Status: Accepted	Only Accepted requests shown		
11-008	Request Management	Search request by Request ID	Request exists	1. Enter Request ID 2. Search	Request ID: BR001	Request details displayed		
11-009	Request Management	Verify status change audit	Request status changed	1. Change status 2. Check audit log	Status: Pending -> Accepted	Audit shows old/new status, staff, timestamp		
11-010	Request Management	Cannot modify fulfilled request	Fulfilled request	1. Try to change Fulfilled status 2. Submit	Status: Fulfilled	Status locked, cannot be changed		

FR-12: Low-Stock Alert

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
12-001	Low-Stock Alert	Alert triggered when below threshold	A+ threshold: 500ml, current: 600ml	1. Remove 150ml 2. Stock becomes 450ml 3. Check alerts	Removed: 150ml, Threshold: 500ml	Alert sent to staff immediately		
12-002	Low-Stock Alert	No alert when stock above threshold	B+ threshold: 500ml, current: 1000ml	1. Remove 300ml 2. Stock: 700ml 3. Check alerts	Removed: 300ml, Final: 700ml	No alert triggered		
12-003	Low-Stock Alert	Alert at exact threshold	O+ threshold: 500ml,	1. Remove 100ml 2.	Removed: 100ml, Final:	Alert triggered at exactly		

			current: 600ml	Stock exactly 500ml	500ml	500ml		
12-004	Low-Stock Alert	Alert includes blood group and quantity	Alert triggered	1. Trigger alert 2. Check alert content	AB+ stock low	Alert shows blood group, current qty, threshold		
12-005	Low-Stock Alert	Only authorized staff receive alerts	Alert triggered	1. Trigger alert 2. Check who received	Blood bank staff notified	Only relevant staff notified		
12-006	Low-Stock Alert	Alerts sent in real-time	Stock update made	1. Update inventory 2. Check alert timing	Update triggers alert	Alert received within 1 minute		
12-007	Low-Stock Alert	Multiple alerts for different groups	Multiple groups low	1. Multiple groups fall below threshold 2. Check alerts	O+, A+, B+ all low	Separate alerts for each group		
12-008	Low-Stock Alert	Alert acknowledged by staff	Alert received	1. Receive alert 2. Click acknowledge 3. Verify	Alert ID: AL001	Alert marked as read, timestamp recorded		
12-009	Low-Stock Alert	Alert resolves when stock increases	Alert active, stock below threshold	1. Add 300ml to low stock 2. Stock above threshold 3. Check alerts	Added: 300ml, Stock restored	Alert status changes to resolved		
12-010	Low-Stock Alert	Verify alert audit trail	Alert triggered	1. Trigger alert 2. Check audit log	Alert for O+	Audit shows trigger time, recipients, acknowledgment		

FR-13: Notifications

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
13-001	Notifications	Notify staff of new request	Request submitted	1. User submits request 2. Check staff notifications	New request for O+	Staff receives notification immediately		
13-002	Notifications	Notify user of request status change	Request status changed	1. Staff updates status to Accepted 2. Check user notifications	Status: Pending -> Accepted	User receives status update notification		
13-003	Notifications	Notify staff of low stock	Stock falls below threshold	1. Inventory update triggers low-stock 2. Check notifications	Stock low for AB-	Staff receives low-stock alert		
13-004	Notifications	Notify donor of donation eligibility	Donor eligible to donate	1. Donation eligibility reached 2. Check donor notifications	Last donation: 56+ days ago	Donor receives eligibility notification		
13-005	Notifications	User customizes notification preferences	User logged in	1. Go to Settings 2. Toggle notifications on/off 3. Save	Disable request alerts	Preferences saved, future alerts follow setting		
13-006	Notifications	Multiple notifications displayed	Multiple events occurred	1. Check notification panel 2. View all notifications	5 different notifications	All notifications listed with timestamps		

13-007	Notifications	Notifications include action links	Notification displayed	1. Receive notification 2. Click link 3. Verify navigation	New request notification	Clicking notification navigates to relevant page		
13-008	Notifications	Notifications sent via email	Email configured	1. Trigger notification event 2. Check email	New request event	Email notification received		
13-009	Notifications	Unauthenticated users receive notifications	Email configured	1. User receives email notification 2. No login required to read	Email notification sent	Notification accessible without login		
13-010	Notifications	Verify notification audit trail	Notification sent	1. Send notification 2. Check audit log	Notification ID: N001	Audit shows recipient, event type, timestamp		

FR-14: Reports

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
14-001	Reports	Generate donor report	Admin logged in	1. Navigate to Reports 2. Select Donor Report 3. Set date range 4. Generate	Date: 2024-01-01 to 2024-12-31	Report showing total donors by blood group, status		
14-002	Reports	Generate donation report	Staff logged in	1. Select Donation Report 2. Filter by date/bank 3. Generate	Date range, Bank filter	Report with donation count, trending, analysis		

14-003	Reports	Generate inventory report	Staff logged in	1. Select Inventory Report 2. Choose blood bank 3. Generate	Bank: Central Bank	Current inventory by blood group, expiry dates		
14-004	Reports	Generate blood request report	Admin logged in	1. Select Request Report 2. Filter by status/urgency 3. Generate	Status: all, Date range	Request count, fulfillment rate, avg time		
14-005	Reports	Export report as PDF	Report generated	1. Generate report 2. Click Export 3. Select PDF	Report type: Donor	PDF file downloaded, formatted correctly		
14-006	Reports	Export report as Excel	Report generated	1. Generate report 2. Click Export 3. Select Excel	Report type: Donation	Excel file with proper formatting and data		
14-007	Reports	Non-authorized user cannot generate	Donor logged in	1. Try to access Reports 2. Generate	User: Donor	Access denied		
14-008	Reports	Report includes charts/graphs	Report generated	1. Generate report 2. Check for visual data	Donation trends	Charts showing trends, comparisons		
14-009	Reports	Schedule automatic report generation	Admin logged in	1. Go to Reports 2. Set schedule 3. Configure	Weekly, email to admin	Report generated and sent automatically		
14-010	Reports	Verify report data accuracy	Report generated	1. Generate report 2. Cross-check	Sample report data	All data matches system records		

				with database				

FR-15: Search and Filtering

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
15-001	Search and Filtering	Search by donor ID	Staff logged in	1. Navigate to Search 2. Enter donor ID 3. Search	Donor ID: D001	Donor details displayed		
15-002	Search and Filtering	Search by donation ID	Staff logged in	1. Search donations 2. Enter donation ID 3. Search	Donation ID: DN001	Donation details displayed		
15-003	Search and Filtering	Filter by blood group	Records exist	1. Select blood group filter 2. Apply	Blood Group: O+	Only O+ records displayed		
15-004	Search and Filtering	Filter by blood bank location	Multiple banks exist	1. Select location filter 2. Apply	Location: Bangalore	Only Bangalore records shown		
15-005	Search and Filtering	Search by request ID	Requests exist	1. Enter request ID 2. Search	Request ID: BR001	Request details displayed		
15-006	Search and Filtering	Combine multiple filters	Records exist	1. Filter O+ blood 2. Filter Bangalore location 3. Apply all	O+ in Bangalore	Only matching records displayed		
15-007	Search and Filtering	Filter by date range	Records exist	1. Select date range 2. Apply	2024-01-01 to 2024-12-31	Only records in range shown		
15-008	Search and Filtering	Clear all filters	Filters applied	1. Click Clear All 2. View results	All filters removed	All records displayed again		

15-009	Search and Filtering	Save filter preferences	Filters configured	1. Set filters 2. Save as preset 3. Apply later	Filter name: Weekly O+	Saved filter can be reused		
15-010	Search and Filtering	Search returns exact matches first	Multiple matches	1. Search by name 2. Check order	Name search: John	Exact match first, then partial matches		

FR-16: Role-Based Access Control

TEST CASE ID	Name of module	Test case description	Pre conditions	Test steps	Test data	Expected results	Actual result	Test result
16-001	Role-Based Access Control	Donor can only access donor functions	Donor logged in	1. Check available menu items 2. Try to access staff functions	User role: Donor	Staff/ Admin functions not visible		
16-002	Role-Based Access Control	Staff can access staff functions	Staff logged in	1. Navigate to inventory 2. Navigate to donations 3. Access staff reports	User role: Staff	All staff functions accessible		
16-003	Role-Based Access Control	Admin can access all functions	Admin logged in	1. Check menu 2. Navigate to admin functions	User role: Admin	All system functions accessible		
16-004	Role-Based Access Control	Staff cannot access Blood Bank Management	Staff logged in	1. Try to manage blood banks 2. Check access denied	User role: Staff	Access denied message shown		
16-005	Role-Based Access	Donor cannot view	Donor logged in	1. Try to access other	Accessing: Other donor	Access denied, can only		

	Control	other donor profiles		donor profile 2. Check access	profile	see own		
16-006	Role-Based Access Control	Staff can only modify assigned bank	Staff1 assigned to Bank1	1. Try to access Bank2 inventory 2. Modify	Staff1 accessing Bank2	Access restricted or denied		
16-007	Role-Based Access Control	Direct URL access without permission denied	Donor logged in	1. Try to access /admin/blood-banks 2. Try /staff/inventory	No permission granted	Redirected to unauthorized page		
16-008	Role-Based Access Control	API endpoint enforces role-based access	Donor user	1. Call API without proper role 2. Make staff-only API call	API: POST /inventory/update	API returns 403 Forbidden		
16-009	Role-Based Access Control	Role change reflects immediately	Admin upgrades user role	1. Change donor to staff role 2. Staff logs in 3. Check menu	Role changed: Donor -> Staff	New staff functions now visible		
16-010	Role-Based Access Control	Verify role-based access audit	User attempted unauthorized access	1. Try unauthorized action 2. Check audit log	Unauthorized access attempt	Audit logs failed access attempts		